

Healthcare Data Science Report

Title:

An Exploratory Data Analysis of National Survey on Drug Use and Health (NSDUH) Data, to inform multi stakeholder decision-making during the United States Opioid Crisis Public Health Emergency.

Prepared by: Student ID

Version: 1.0

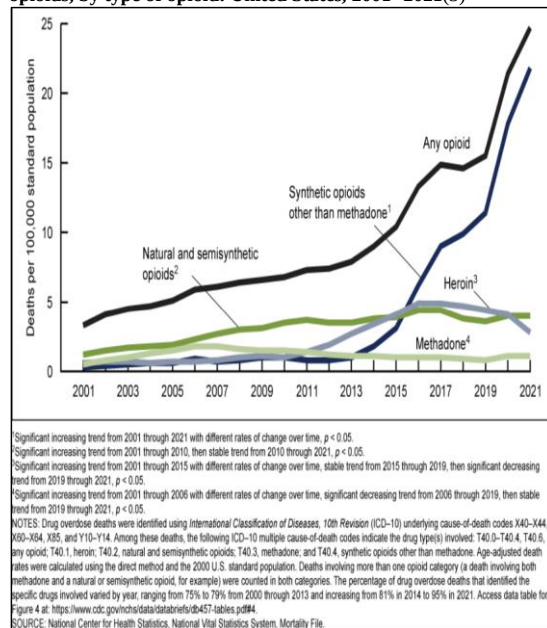
Created: 26/Mar/2023

1.1. Motivation / Importance of the Data

In 2018, the Director of the Center for Disease Control (1) attributed America's drop in life expectancy to deaths from drug overdose and suicide (1). The top ten causes of death between 2016 to 2017 had just been reported: Heart disease, Cancer, Unintentional injuries and Suicide featured 1st, 2nd, 3rd and 10th respectively(2). Death by Unintentional injury is an umbrella category that includes drug overdoses and together with suicides, had begun to erode decades of gains the USA previously enjoyed for life expectancy.

Commented [JK1]: Excellent motivation!

Figure 1: Age-adjusted rate of drug overdose deaths involving opioids, by type of opioid: United States, 2001- 2021(3)



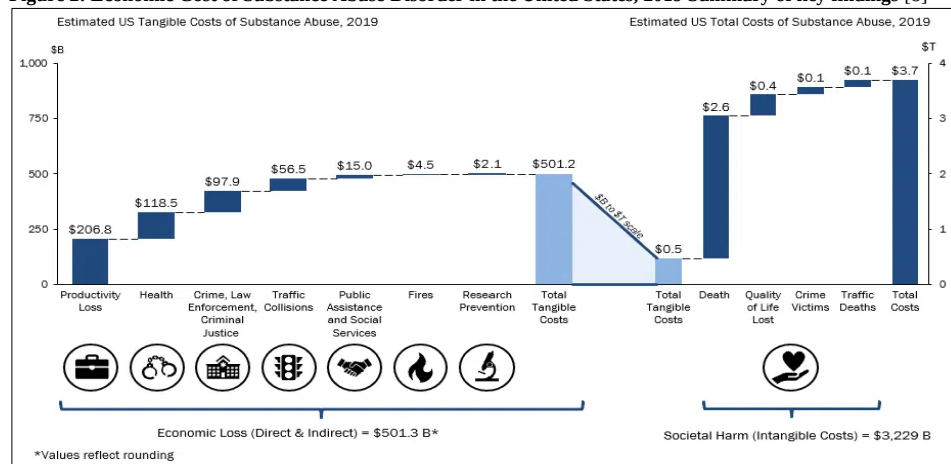
Whilst opioid overdose deaths accounted for 70.6% of all drug overdose deaths in 2019 (Figure 1), other trends in this category were developing. From 2012 through to 2019, the rate of drug overdose deaths involving cocaine increased more than 3-fold (from 1.4 to 4.9), while those involving psychostimulants with abuse potential increased more than 6-fold (from 0.8 to 5.0)(4). The Surgeons General Report: Facing Addiction in America September 2018(5) cast a spotlight on the devastating effects in particular of opioid misuse and overdose. Notwithstanding premature death, opioid misuse can result in:

- adverse health conditions in the user (such as cardiovascular abnormalities(6) and exposure to infectious diseases(7))
- complications during and after pregnancy(8)
- enormous direct and indirect economic & social costs

An Economic Cost Analysis published in April 2020 tried to quantify tangible direct costs such as health expenditure, and indirect tangible costs (such as loss of productivity from premature death, absenteeism or incarceration) that substance abuse has on American Society; estimating these to amount to \$0.5 Trillion USD(9) in 2019 alone (Figure 2).

Commented [JK2]: Strong back up to the importance

Figure 2: Economic Cost of Substance Abuse Disorder in the United States, 2019 Summary of key findings [8]





With the opioid crisis declared a Public Health Emergency(10) and the National Institute of Health investing an initial \$2 billion USD into its Helping to End Addiction Long-term Initiative(11), it is logical to want to understand the prevalence and patterns of substance misuse; in order to provide support where it is most needed, and mitigate the heaviest burdens of substance misuse on society.

The National Survey on Drug Use and Health (NSDUH) is conducted annually in the USA via face-to-face, computer-assisted self-interviewing and (since 2020) web-based interviews, with persons 12yrs and older. Surveys are conducted 4 times a year and follow a sample size design(12) to obtain a proportional representation of responses from each state. In 2019, 67,625 interviews were completed.

As the NSDUH data provides estimates of substance use and mental illness across the USA, this dataset can be explored to reveal annual patterns and trends of substance misuse. The impacts of substance misuse are not homogenous across the population; the NSDUH dataset can show how age groups, genders, racial backgrounds, rural vs urban communities are affected differently. The substances that are misused or abused also vary over time and place - the dynamic nature of this opioid crisis is estimated to have moved through at least 3 overlapping phases by 2019(13). The NSDUH could offer insights into which substances of abuse are experiencing transient notoriety vs deep rooted persistence.

It is on this basis that the NSDUH dataset can be considered an informative resource for both government and non-governmental stakeholders to aid decision making and resource allocation, such as drug specific treatment services and prevention programs tailored to the needs of their population.

1.2. Stakeholders

With a health problem that can be simultaneously insidious and crystal clear, it is difficult to think of a stakeholder who would have no interest in leveraging the NSDUH dataset. The following are key stakeholders who would have interest in at least one step of the project pipeline (Figure 3).

The CDC: utilizes NSDUH for UpToDate information on tobacco, alcohol and drug use in the United States. They contribute greatly towards the US Department of Health and Human Services, shaping health promotion, prevention and preparedness policies; and would have an interest in:

- how the data is collected and curated
- any trends and problems that this dataset can identify
- whether accurate forecasting models can be created
- disseminating the findings and/or modelling to other public health stakeholders

Secondary care setting: community hospitals (private/not for profit, government run) would want to:

- understand how the data is curated and is applicable to their population
- utilize the trends from this data and possible modelling to ensure adequate service provision; for example of their Emergency Departments, counselling or psychotherapy teams, crisis planning, and upskilling of their staff

Researchers and Research Funding Agencies: these stakeholders would use this data to research any risk and/or protective factors to substance misuse. They may look for any measure of the effectiveness of a particular policy decision/ prevention campaign/ treatment service through temporal association.

Private and Public Health Insurers: in response to recent requests to reshape their policies to ameliorate the opioid pandemic(14); insurers would look to this data to help them:

- identify patterns of substance misuse (particularly of opioid prescription drugs for acute pain)
- applying these learnings to identify misuse in their policy holders
- design better and more competitive insurance packages, by way of offering their policy holders timely connection or referral to pain management services and addiction treatment services.

Society and Patients: Besides fiscal impacts, Society stands to gain a lot from analysis of this data, provided the information is shared to all levels of health literacy. This may come direct from the media, but also via all the stakeholders listed above. Patients suffering from substance misuse would have access to services that meet their needs, enabling them to recover from their addiction.

Pharmaceutical Industry: whilst no mea culpa has ever been offered, the pharmaceutical industry would have a role in alleviating the opioid crisis. Analysis of this data set could:

- enable focused drug development of new, more acceptable and effective treatments,
- give a better understanding of how prescription opioids that are therapeutically beneficial when used appropriately, are being misused

Commented [JK3]: Good to include this important information

Commented [JK4]: Not clear how the data can show this

Commented [JK5R4]: Its an interesting question

Commented [JK6]: Excellent - Agreed that there are value to many stakeholders

Appears to cover all the major stakeholders

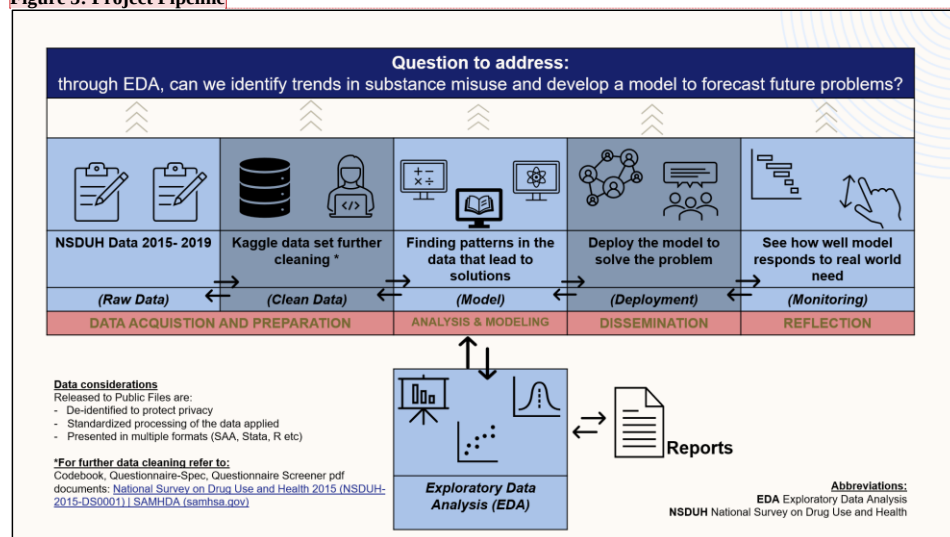
1.3. Project pipeline

Figure 3 describes the proposed project pipeline. The Kaggle dataset(15) for this project was derived by merging 2015-2019 Public Use Files (PUF) available from the Substance Abuse & Mental Health Data Archives (SAMHDA); meaning that it is de-identified and all variables, missing data codes and coding scheme are standardized(16).

To ensure correct understanding of each variable code, the SAMHDA code book will need to be consulted and understood (16). From time to time, data variables are redefined or introduced via modifications to the survey. To make comparisons between certain variables, Variable Crosswalk Charts are available (16). The NSDUH underwent partial redesign in 2015, rendering pooling of some variable data before 2015 and after 2015 not possible(17) (17). This does not impact this analysis given the pooling is between 2015 and 2019. Finally, due to significant changes in data collection methodology and not achieving a full year of data collection during the covid19 pandemic, 2020 data is considered a non-comparable year to 2019 nor 2021 (16).

Through each stage of the pipeline, it will be important to assess how robustly the project still addresses the 'Project Question'. Consequently, every step may generate refinements that are applied to precedent steps.

Figure 3: Project Pipeline



Commented [JK7]: great

Commented [JK8]: Good use of a figure to highlight the pipeline - the data is interesting and will lead to potentially important findings. As a Kaggle dataset, it is already available and ready for analysis

1.4. Expected Outcomes

As survey responders are not followed over time, each year's dataset is cross-sectional rather than longitudinal. However, through Exploratory Data Analysis of the variables over the years, substance use patterns and trends may emerge, for example:

- Age of first substance use
- Prevalence by age group, gender, racial backgrounds or other socioeconomic descriptors
- Most abused substances
- Any associated mental health issues
- Utilization of prevention and treatment programs

Through Data Visualization and Descriptive Statistics delivered by this project, Stakeholders will be enabled to make well informed decisions - both in the immediate and long term- in an agile and concerted way. This may include:

- determining what research is needed in order to build evidence based and effective public health campaigns, treatment guidelines or models of care
- test through modelling, which interventions have the most profound diminishing effect on the worst outcomes that substance misuse has on society
- plan, adjust and deploy front line healthcare resources such as clinics, specialists and other allied health that meet the needs of the population

Further, if risk factors to escalating abuse of substances can be identified, it may be possible to predict by machine learning, future risk of misuse or abuse. This would be of value particularly in early intervention or prevention campaigns.

Commented [JK9]: Good - keep in mind that data sampling will influence these results alot

Commented [JK10R9]: Doesn't answer the Q: "experiencing transient notoriety vs deep rooted persistence"

Commented [JK11]: This will be very interesting to explore - there maybe patterns that could lead to a useful prediction model

2.1. Identify and justify what AI / machine learning methods you will adopt

As this dataset is cross-sectional and population based, a Machine Learning (ML) model that predicts risk of developing OUD at an individual patient level is not possible and would be highly erroneous. Instead, this project will attempt to deliver a ML Model that is capable of accurately predicting the risk - at the population level – of people 12yrs and older developing opioid use disorder (OUD).

Previous studies have sought to determine risk factors for OUD based on conventional statistical methods(18, 19) like multivariable logistic regression, and there are at least seven validated tools/questionnaires that a clinician can use to predict the risk of divergent drug related behaviours or prescription opioid misuse in conditions such as chronic pain (20). With more than 2800 column variables and 280,000 responders in this dataset, this is a reasonably complex, high-dimensional dataset which conventional statistical methods for predictions may not perform as well on (21). EDA may reinforce our understanding of variables that are widely accepted as risk factors to OUD but might also uncover other variables that have a predictive relationship on developing OUD. Multicollinearity is likely to exist within this dataset and understanding these predictor variables (strong positive or negative correlation) will lead to the appropriate engineering of 'features' for training a predictive ML model.

A literature search was conducted to learn from previous studies that leveraged ML models to predict OUD. After searches in databases that have a high degree of pre-appraised evidence yielded no results (Global International Network Database and NICE guidelines), a search for systematic reviews was undertaken in Pubmed based on the below PICO question (Table 1), as systematic reviews yield the next best degree of pre-appraised evidence. One systematic review was yielded by this search (22) which was a systematic review of Geographic Information Science methodologies to understand geographical patterns in opioid related health measures. In the absence of other systematic reviews, primary studies were searched. Three highly relevant studies provided great insight into how to leverage ML models for predicting OUD: (23), (24) and (25).

Table 1: PICO question to yield examples of Machine Learning Models tested to predict Opioid Use Disorder.

P	Opioid Use Disorder
I	Machine Learning
C	Conventional prediction methods (conventional statistical analysis and or/validated clinical tools)
O	Sensitivity and specificity (accuracy) for predicting OUD

In one study(20), a novel Ensemble ML model to prospectively predict individual OUD cases was developed by training the models on a representative Canadian sample. Canada has a centralized, universal and publicly funded health care system which is in stark contrast to the USA. Further, this study had ethics approval to link seven different databases at a patient level, providing a very rich dataset to train for individual risk prediction. With an initial 62 features identified in their dataset (informed by literature search and curated by human experts) the study showed what the top 10 predictors to OUD were in their dataset. Whilst they did not specify exactly which ML models they used to create the ensemble ML model; they did confirm decisions trees, gradient boosted trees and deep neural network learning models were employed, and voting used to combine the results of the individual models to obtain a final prediction.

A second study (22) selected a logistic regression ML model that was trained and validated on Medicaid data, on the basis that of all non-elderly adults diagnosed with an OUD, 38% rely on Medicaid for health insurance(26) therefore suggesting Medicaid beneficiaries are disproportionately affected by OUD. The model to predict future OUD diagnosis had a high specificity, however the sensitive was low because of low prevalence of OUD in the dataset.

Finally, the third study utilized 2014-2017 NSDUH dataset split 70% to 30% into training set and test set respectively, to train 3 ML models (artificial neural networks, distributed random forest, and gradient boosting machine) to predict OUD in adolescents. The ML models were then fitted using 5-fold cross validation and their performance compared against the performance of the logistic regression in a nationally representative sample of U.S. adolescents ages 12–17 years(24). The distributed random forest was deemed as the best-performing model. Three variables (marijuana use experience, tobacco use experience and alcohol use experience) were found to be the most influential variables in explaining opioid misuse in adolescents across all 3 machine learning models.

Table 2: from ‘Using machine learning to predict opioid misuse among U.S. adolescents’ (24)

Table 2. Performance results of three different machine learning models and the penalized logistic regression in predicting opioid misuse among U.S. adolescents.		
Model	AUROC	AUPRC
Artificial neural networks (threshold=0.196)	0.810	0.157
Distributed random forest (threshold=0.114)	0.809	0.172
Gradient boosting machine (threshold=0.055)	0.811	0.160
Penalized logistic regression (threshold=0.139)	0.815	0.162

Note: AUROC - area under the receiver operating characteristic curve, AUPRC - area under the precision-recall curve. Values in the parentheses represent the optimized probability thresholds that maximize the accuracy.

As this project aims to predict something from a dataset that contains a variety of continuous, discrete, ordinal, and categorical data, most of which has been transformed to numerical data by means of a code data dictionary, a supervised ML model should be fit for purpose. Distributed Random Forest machine learning model is supervised, can handle a large number of features and has flexibility and accuracy beyond standalone decision trees when it is given new data to classify.

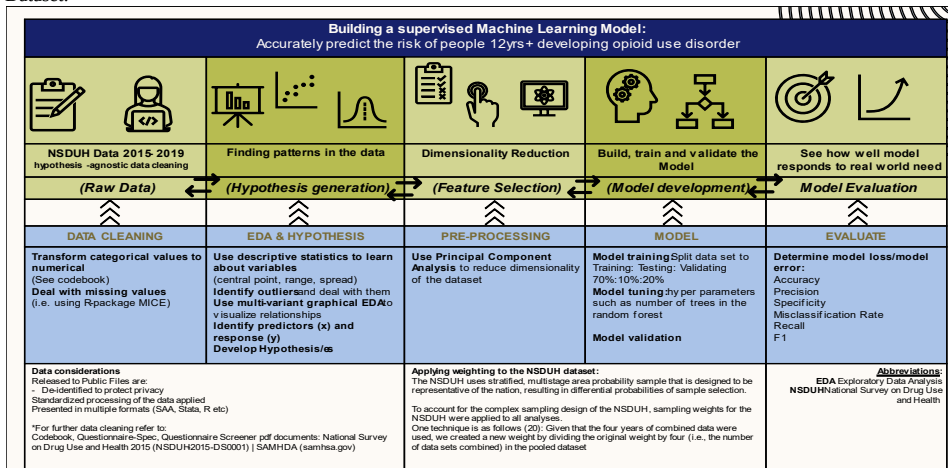
Based on this literature review, the type of data in the NSDUH dataset and the predictive aspect this project hopes to achieve, the random forest model is chosen to explore.

2.2. Methods for Developing the proposed model

Figure 4 is a top-level overview of developing a supervised distributed random forest machine learning model for predicting Opioid Use Disorder from the NSDUH Dataset.

No Human Research Ethics Committee approval need to be obtained for this EDA and ML model development, as per section 1.3 of this report, the Kaggle dataset (15) was derived by merging 2015-2019 Public Use Files (PUF) available from SAMHDA; meaning that it is de-identified and already in csv format for reading into R. It is important to state that throughout the development of this proposed model (and indeed throughout the model’s deployment), the USA AI Risk Management Framework(27) akin to Australia’s Artificial Intelligence Ethics Framework (28) will be taken into consideration so as to foster trust and responsible use once deployed.

Figure 4: Method for developing supervised machine learning model for predicting Opioid Use Disorder from NSDUH Dataset.



Step 1: Data cleaning

Some preliminary EDA has been performed on this data set. The data is structured, and categorical data values have been transformed to numerical values as per the coding laid out in the data dictionary (16). Continuous data columns tend to use three-digit special codes, whereas ordinal and categorical variables use two-digit codes. Although all the variables, missing data codes and coding scheme have been standardized, there are codes that are equivalent to N/A. These may need to be imputed, and if so known techniques in R such as MICE – multiple imputations with chain equations – will be utilized.

Step 2: EDA, Feature Selection and Hypothesis generation

With consideration to defining the target/ source/ study population, the target population is anyone in the USA 12yrs or over. The source population is responders to the NSDUH survey between 2015 – 2019. The study population is the subset of NSDUH survey responders who indicated that they have misused or abused pain relievers or heroine anytime in the last 12 months and meet the definition of OUD.

Guided by the literature review conducted earlier, some exploration will be undertaken to learn more about the independent variables which are potential predictors (x) of the outcome (y). The outcome variable (y) needs to meet the definition of Opioid Use Disorder: dependence or abuse of prescription opioid pain relievers or use of heroin (29).

As such the outcome variable (y) may be a newly derived binary variable (0=no, 1=yes), based on responders confirming they misuse any one of the available prescription opioid pain relievers to select from the survey (i.e. buprenorphine, oxycodone, morphine) or heroin in the last 12 months, or did not.

Descriptive statistics and data visualization will be used to understand any patterns, look at the distribution and spread of that variable or interesting relations among variables. Notably, EDA will look for correlations in the data which can be represented in scatter plots. Given there are over 2800 column variables in this data set, the risk of the ‘Curse of Dimensionality’ impacting the ML model is high, particularly as Random Forest algorithms are used for both classification and regression tasks. The literature review conducted earlier also supports the possibility that there could be multiple independent variables that have an exerting effect on the outcome. It may be worthwhile to perform some dimensional reduction in this dataset using the Principal Component Analysis, thereby understanding which variables have the biggest impact on the outcome variable and guide on the features needed to accurately train the model on.

Step 3: Training the model

The NSDUH dataset is quite large and can be split into training and testing datasets 70% and 30% respectively. To mitigate overfitting, the training dataset could be shuffled and further split as per a cross-validation method such as K-fold. The validating dataset will be used to tune any hyperparameters, before the test dataset set is used to evaluate the models hyperparameters generalizability and subsequent performance at predicting OUD.

2.3. Approach to Evaluating the model

Given that the NSDUH survey is so standardized, it would in theory be possible to give the model new data from beyond 2019. However it has been noted that due to the covid19 pandemic, in 2020 only 2 quarters (Q1 and Q4) had any data collection, and in 2021 the NSDUH data collection methods varied and included web interviews, therefore advising against comparing 2020 and 2021 data to prior years(30).

Nevertheless, there are several ways to evaluate the model. As we have our y outcomes, one of the ways to evaluate the model is using a confusion matrix as per figure 5 below.

Figure 5: example confusion matrix to evaluate Random Forest Model performance.

		Actual	
		OUD	No OUD
Predicted	OUD	True Positive (TP)	False Positive (FP)
	No OUD	False Negative(FN)	True Negative (TN)

From here, a range of metrics can be calculated as follows(31):

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total Predictions}}$$

How much the OUD Prediction model classifies the right output (either yes OUD, or No OUD)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Proportion of actual OUD that the prediction model detected



Specificity=	$\frac{TN}{TN+FP}$	Proportion of real negatives
Error Rate =	$\frac{FP+FN}{\text{Total Predictions}}$	How much the predictor model classifies wrongly
Recall=	$\frac{TP}{TP+FN}$	How well the model predicts the 'yes OUD' when it is actually yes OUD.
F1 score=	$\frac{2 \times \text{Precision} \times \text{recall}}{\text{Precision} + \text{recall}}$	Defined as the weighted average of the actual positive levels (recall) and consistency. F1 is a useful general measure of a model's precision that combines accuracy with recall.

A Receiver Operating Characteristic (ROC) curve (24) is a graphical means of evaluating the predictor model. Area Under the ROC and Area Under the precision-recall curve may be useful to visualize (24). AUPRC plots positive predicted values against true positive rates (32) and has been considered more informative for assessing binary classified on imbalance outcome variable (33), which is likely in this NSDUH dataset.

2.4. Addressing Stakeholder's interests

The CDC: deploying this ML model into the NSDUH database would help the CDC rapidly understand what the trends and risks are for 12yr old+ population developing opioid use disorder (OUD). The CDC contribute greatly towards the US Department of Health and Human Services, shaping health promotion, prevention and preparedness policies; and dissemination of these latest insights would encourage appropriate use of resources to limit OUD.

Secondary care setting: community hospitals (private/not for profit, government run) would utilize the findings from this ML model to adjust their approach to identifying potentially at-risk patients; both in the ambulatory setting and Emergency Departments.

Due to time constraints of this project, this dataset is not linked to anything else and so cannot predict on an individual patient level, risk of developing OUD. Ultimately, if the project learnings could be applied to a highly linked dataset (i.e. human services and criminal justice data, insurance claims databases, pharmacy prescription dispensing data, electronic health records data) then expensive and time-consuming questionnaire based tools used in clinics to quantify risk, could be replaced with a ML model; with Healthcare Professionals confident to act on it.

Researchers and Research Funding Agencies: these stakeholders would use the outcomes of this ML model to test effectiveness of a particular policy decision/ prevention campaign/ treatment service through temporal association.

Private and Public Health Insurers: insurers would look to this ML model to inform them on how to adjust their surveillance of their own policy holders to identify potential OUD. They can continually assess and improve insurance packages, by way of offering their policy holders timely connection or referral to and addiction treatment services.

Society and Patients: Besides fiscal impacts, Society stands to gain a lot from analysis of this data, provided the information is shared to all levels of health literacy. This may come direct from the media, but also via all the stakeholders listed above. Patients suffering from substance misuse would have access to services that meet their needs, enabling them to recover from their addiction.

Pharmaceutical Industry: would find the insights from this ML model beneficial to understanding of how prescription pain relievers that are therapeutically beneficial when used appropriately, are being misused.

3.1. Experimental design and details

3.1.1. Packages

As a complex dataset of survey data with more than 2800 column variables and more than 280,000 responders, it was important to install packages that would aid pre-processing and data cleaning.

- **tidyverse** was selected for its multiple functions like readr for data import, ggplot2 for data visualization, dplyr for data manipulation and tibble for re-imagining dataframes.
- **zoo** was selected to help handle missing values. In particular, the code utilizes the na.aggregate() function from the "zoo" package to impute missing values in the dataset. Imputing missing values with column medians rather than mean, provided a simple and effective way to fill in the gaps in the data, allowing for more complete and reliable analysis.
- **survey** was useful for analysis of complex survey samples
- **gtsummary** was a useful package for assisting with data summary and analytic results tables
- **caret**, **randomForest** was used for pca and prediction model training

3.1.2 New csv files

At various stages, new csv files were created by calling up the write.csv function.

- **Fullimputed_NSDUH.csv** : this had all NAs imputed to median value of column and had the survey weight applied. This file was saved off incase for Machine Learning I would use the entire dataset
- **FinalCleanedNSDUH.csv**: this is just the variables of interest, median imputed NAs and weighted with an new ID column created (total of 54 columns)

3.1.3 Extracting columns of interest into new dataframe called NSDUH Subset

A closer read of the 2019 NSDUH codebook revealed that a new column variable for opioid use disorder was not required (refer to section 2.2 Step 2). The column variable udpyopi is a recoded column in the dataset whereby if the response to any of these four variables is '1=yes' then the responder is automatically described as having OUD:

- **depndher**: responded to questions that mean that met 3 or more of the 7 DSM-IV criteria for heroin dependence
- **depndpyprn**: responded to questions that mean that met 3 or more of the 7 DSM-IV criteria for prescription pain relieve dependence
- **abuseher**: response to questions did not meet DSM-IV criteria for dependence, but they reported one or more of the 3 criteria for abuse of the substance
- **abusepyprn**: response to questions did not meet DSM-IV criteria for dependence, but they reported one or more of the 3 criteria for abuse of the substance

Consulting the 2019 NSDUH codebook (16) and aligning with the literature review performed, groups of variables were extracted from the dataset for closer EDA and ML model building (Table 1). This formed a dataset of 54 column variables, more than 280,000 rows, with NAs fully imputed with medians and survey weighting applied.

3.1.4: PCA, model training and evaluation

This subset was split 80:20 train and test using the caret package in R. There were several column variables in this dataset that directly feed into the y variable "udpyopi" (Opioid Use Disorder). If the response to any of these four variables is 'yes' then they are automatically described as OUD: depndher, depndpyprn, abuseher, abusepyprn. So when training and testing, these four columns were avoided, along with the year and the id column – refer to figure 6. Then PCA was applied to the input(independent) variables. A multilinear regression model was then trained with just the first nine principle components, with a 10 fold cross validation to prevent overfitting, and the model performance assessment conducted. Because the cumulative proportion was only at 37% by PC9, it was decided to re-run but up to PC36, which was 89% culmulative proportion by that point.

The next steps of model training were to train and test the data on other machine learning models that are more appropriate for classification, particularly as the y variable is binary and categorical (yes or no), and the dataset itself was also found to be extremely skewed. The classification models that were selected were Gaussian Naïve Bayes, Bernoulli Naïve Bayes, Decision Tree Classifier, Random Forest and Support Vector Classification. All of these models were then evaluated by creating confusion matrices and outputting Accuracy, Precisions, Recall, Specificity, Error rate and F1 scores. These were done in google colab using pandas and sklearn packages.

Table 1: Columns of Interest extracted for data cleaning and prediction model training

Theme/Type of variable	Column name	Independent (x) or dependent (y) variable
Demographic/socioeconomic/geographic: (10 variables)	"irsex", "catag6", "eduhighcat", "newrace2", "irwrkstat", "income", "irinsur4", "govtprog", "coutyp4", "year"	x
Other Health variables: (5 variables)	"health2", "amhinp2", "amiyr_u", "amdelt", "amdeyr".	x
Past 12 months prior Substance Abuse variables: (12 variables)	"abusepypnr", "abusealc", "abusemrj", "abusecoc", "abuseher", "abusepyhal", "abusepyinh", "abusepymth", "abusepytrq", "abusepystm", "abusepysed", "abusepypsy".	x
Prior 12 months Substance Dependence variables: (13 variables)	"depndalc", "depndmrj", "depndcoc", "depndher", "depndpyhal", "depndpyinh", "depndpymth", "depndpypnr", "depndpytrq", "depndpystm", "depndpysed", "depndpypsy", "ndssdnsp".	x
Reasons given for misusing Prescription Pain Relievers variables: (8 variables)	"pnrrsrelx", "pnrrshigh", "pnrrsdept", "pnrrspain", "pnrrsemo", "pnrrsdgdx", "pnrrshook", "pnrrslep".	x
Y variable:	UDPYOPI	y
Other columns for survey weighting	verep, vestr, analwt_c	Survey weights
Column of sequential id numbers	id	id
<i>Refer to appendix C for full summary of responses to each variable</i>		

Figure 6: Dimensionality Reduction

```
# Step 2: Split the REALLY cleaned up dataset (NSDUH_PCA is just the 54 columns of interest, with survey weight and median imputed NAs) into
train and test sets. Repeat all of these steps on the imputed_NSDUH dataset which is 2812 columns, with survey weight and median imputed
NAs)
set.seed(42)
trainset <- NSDUH_PCA %>% dplyr::sample_frac(0.80)
testset <- dplyr::anti_join(NSDUH_PCA, trainset, by = 'id')
...[r]
# Step 3: select and scale independent variables and the dependent variable, I have removed several including year, id, udgpyopi(y variable)
and abuseher, abusepypnr, depndher, depndpyp, depndpypnr because if either of them score 1, then person is OUD (i.e. udgpyopi is 1)
trainX <- select(trainset, ~c(year, udgpyopi, abuseher, abusepypnr, depndher, depndpypnr, id))
testX <- select(testset, ~c(year, udgpyopi, abuseher, abusepypnr, depndher, depndpypnr, id))
xPreprocessor <- preProcess(trainX, method = c('center', 'scale'))
normTrainX <- predict(xPreprocessor, trainX)
normTestX <- predict(xPreprocessor, testX)
trainY <- select(trainset, c(udgpyopi))
testY <- select(testset, c(udgpyopi))
yPreprocessor <- preProcess(trainY, method = c('center', 'scale'))
normTrainY <- predict(yPreprocessor, trainY)
normTestY <- predict(yPreprocessor, testY)
```

3.2. Results

3.2.1 Summary Statistics

After successfully cleaning the NSDUH dataset, a series of bar charts were created, showing the spread of each variable. However, it was more helpful to display relationships through means of creating large summary statistics tables. These tables additionally had p value calculated by Pearsons Chi-squared test. This was appropriate because the data set is heavily skewed towards not having OUD: 0.89% of those surveyed had OUD (2514 vs 280254), and we are determining a relationship between two types of categorical data.

The following series of summary statistics were created in R studio.

Table 2: Socioeconomic, Demographic and geographic variables vs Opioid Use Disorder status, stratified by gender

Table 3: what other substances were abused by the responders vs Opioid Use Disorder status, stratified by gender

Table 4: what other substances were responders dependent on vs Opioid Use Disorder status, stratified by gender

Table 5: General health and mental health descriptors vs Opioid Use Disorder status, stratified by gender

Table 6: Reasons for using prescription pain relievers not as directed by the doctor vs Opioid Use Disorder status, stratified by gender

Table 2: Socioeconomic, Demographic and geographic variables vs Opioid Use Disorder status, stratified by gender

Characteristic	No Opiod Use Disorder, N=280254			Opioid Use Disorder, N=2514		
	Male N =	Female N =	p-value ²	Male N =	Female N =	p-value ²
	133,367 ¹	146,887 ¹		1,290 ¹	1,224 ¹	
Age bracket			<0.001			<0.001
12-17 yrs old	34,776 (26%)	33,156 (23%)		112 (8.7%)	219 (18%)	
18-25 yrs old	33,322 (25%)	35,794 (24%)		417 (32%)	383 (31%)	
26-34 yrs old	19,873 (15%)	23,513 (16%)		346 (27%)	284 (23%)	
35-49 yrs old	25,443 (19%)	30,579 (21%)		297 (23%)	247 (20%)	
50-64 yrs old	11,506 (8.6%)	13,529 (9.2%)		99 (7.7%)	79 (6.5%)	
65yrs or older	8,447 (6.3%)	10,316 (7.0%)		19 (1.5%)	12 (1.0%)	
Race			<0.001			0.22
White	78,296 (59%)	84,910 (58%)		873 (68%)	817 (67%)	
Black/African American	16,306 (12%)	19,577 (13%)		113 (8.8%)	102 (8.3%)	
Native Amer/Alaska Native	1,966 (1.5%)	2,087 (1.4%)		26 (2.0%)	36 (2.9%)	
Native Hawaii/Other Pac Isl	707 (0.5%)	713 (0.5%)		6 (0.5%)	8 (0.7%)	
Asian	6,257 (4.7%)	6,682 (4.5%)		23 (1.8%)	14 (1.1%)	
Multi race	5,144 (3.9%)	5,574 (3.8%)		59 (4.6%)	77 (6.3%)	
Hispanic	24,691 (19%)	27,344 (19%)		190 (15%)	170 (14%)	
Education Level			<0.001			<0.001
Less than Highschool	13,983 (10%)	13,456 (9.2%)		259 (20%)	157 (13%)	
High School Grad	28,725 (22%)	27,567 (19%)		434 (34%)	332 (27%)	
Some College/Assoc Dg	30,702 (23%)	40,491 (28%)		389 (30%)	399 (33%)	
College Grad	25,181 (19%)	32,217 (22%)		96 (7.4%)	117 (9.6%)	
12-17 yrs old	34,776 (26%)	33,156 (23%)		112 (8.7%)	219 (18%)	
Employment Status			<0.001			<0.001
Employed Full Time	59,956 (45%)	52,548 (36%)		539 (42%)	336 (27%)	
Employed Part time	16,455 (12%)	25,734 (18%)		163 (13%)	211 (17%)	
Unemployed	8,665 (6.5%)	7,770 (5.3%)		230 (18%)	158 (13%)	
Other(inc.not in labor force)	31,202 (23%)	44,646 (30%)		333 (26%)	447 (37%)	
12-14yrs old	17,089 (13%)	16,189 (11%)		25 (1.9%)	72 (5.9%)	
Total Family Income			<0.001			<0.001
Less than \$20k USD	22,723 (17%)	30,351 (21%)		347 (27%)	423 (35%)	
\$20k-\$49,999	39,871 (30%)	45,515 (31%)		441 (34%)	420 (34%)	
\$50k - \$74,999	21,176 (16%)	22,291 (15%)		197 (15%)	156 (13%)	
\$75k or more	49,597 (37%)	48,730 (33%)		305 (24%)	225 (18%)	
Health Insurance			<0.001			<0.001
Yes	118,414 (89%)	134,396 (91%)		1,028 (80%)	1,066 (87%)	
No	14,953 (11%)	12,491 (8.5%)		262 (20%)	158 (13%)	
Government Assistance			<0.001			<0.001
Yes	25,304 (19%)	34,781 (24%)		466 (36%)	587 (48%)	
No	108,063 (81%)	112,106 (76%)		824 (64%)	637 (52%)	
Rural to urban Continuum			0.63			0.43
Large Metro	60,201 (45%)	66,314 (45%)		553 (43%)	494 (40%)	
Small Metro	46,832 (35%)	51,762 (35%)		458 (36%)	450 (37%)	
Non-metro	26,334 (20%)	28,811 (20%)		279 (22%)	280 (23%)	
¹ n (%)						
² Pearson's Chi-squared test						

Table 3: what other substances were abused by the responders vs Opioid Use Disorder status, stratified by gender

	No Opioid Use Disorder, N=280254			Opioid Use Disorder, N=2514		
	Male, N =	Female, N =	p-value ²	Male, N =	Female, N =	p-value ²
Abuse of other substance in past 12 months	133,367 ¹	146,887 ¹		1,290 ¹	1,224 ¹	
Alcohol	5,076 (3.8%)	3,277 (2.2%)	<0.001	155 (12%)	105 (8.6%)	0.005
Marijuana	1,420 (1.1%)	862 (0.6%)	<0.001	98 (7.6%)	51 (4.2%)	<0.001
Cocaine	162 (0.1%)	101 (<0.1%)	<0.001	51 (4.0%)	29 (2.4%)	0.024
Heroin	0 (0%)	0 (0%)		26 (2.0%)	21 (1.7%)	0.58
Hallucinogens	156 (0.1%)	101 (<0.1%)	<0.001	38 (2.9%)	32 (2.6%)	0.61
Inhalants	66 (<0.1%)	52 (<0.1%)	0.069	12 (0.9%)	18 (1.5%)	0.21
Methamphetamines	88 (<0.1%)	47 (<0.1%)	<0.001	27 (2.1%)	21 (1.7%)	0.49
Tranquilisers	132 (<0.1%)	140 (<0.1%)	0.76	74 (5.7%)	53 (4.3%)	0.11
Stimulants	67 (<0.1%)	57 (<0.1%)	0.15	24 (1.9%)	31 (2.5%)	0.25
Sedatives	14 (<0.1%)	23 (<0.1%)	0.23	10 (0.8%)	14 (1.1%)	0.34
Psychotherapeutic	196 (0.1%)	202 (0.1%)	0.51	235 (18%)	215 (18%)	0.67
Pain Reliever	0 (0%)	0 (0%)		244 (19%)	224 (18%)	0.69
¹ n (%)						
² Pearson's Chi-squared test						

Table 4: what other substances were responders dependent on vs Opioid Use Disorder status, stratified by gender

	No Opioid Use Disorder, N=280254			Opioid Use Disorder, N=2514		
	1, N = 133,367 ¹	2, N = 146,887 ¹	p-value ²	1, N = 1,290 ¹	2, N = 1,224 ¹	p-value ²
Other substance dependence in past 12 months						
Alcohol	4,666 (3.5%)	3,866 (2.6%)	<0.001	257 (20%)	226 (18%)	0.35
Marijuana	2,564 (1.9%)	1,754 (1.2%)	<0.001	180 (14%)	149 (12%)	0.19
Cocaine	289 (0.2%)	204 (0.1%)	<0.001	132 (10%)	99 (8.1%)	0.063
Heroin	0 (0%)	0 (0%)		392 (30%)	285 (23%)	<0.001
Hallucinogens	98 (<0.1%)	60 (<0.1%)	<0.001	37 (2.9%)	21 (1.7%)	0.054
Inhalants	28 (<0.1%)	40 (<0.1%)	0.29	14 (1.1%)	11 (0.9%)	0.64
Methamphetamines	353 (0.3%)	288 (0.2%)	<0.001	140 (11%)	107 (8.7%)	0.076
Tranquilisers	116 (<0.1%)	137 (<0.1%)	0.58	121 (9.4%)	130 (11%)	0.3
Stimulants	225 (0.2%)	232 (0.2%)	0.48	83 (6.4%)	88 (7.2%)	0.45
Sedatives	32 (<0.1%)	42 (<0.1%)	0.45	27 (2.1%)	37 (3.0%)	0.14
Psychotherapeutic	347 (0.3%)	387 (0.3%)	0.87	823 (64%)	841 (69%)	0.009
Pain Reliever	0 (0%)	0 (0%)		797 (62%)	810 (66%)	0.022
Nicotine	8,382 (6.3%)	8,640 (5.9%)	<0.001	560 (43%)	499 (41%)	0.18
¹ n (%)						
² Pearson's Chi-squared test						

Table 5: General health and mental health descriptors vs Opioid Use Disorder status, stratified by gender

	No Opioid Use Disorder, N=280254			Opioid Use Disorder, N=2514		
Characteristic	Male N =	Female N =	p-value ²	Male N =	Female N =	p-value ²
Overall Health Description	133,367 ¹	146,887 ¹	<0.001	1,290 ¹	1,224 ¹	0.017
Excellent	34,523 (26%)	37,375 (25%)	<0.001	121 (9.4%)	98 (8.0%)	0.3
Very Good	51,774 (39%)	56,113 (38%)		407 (32%)	361 (29%)	
Good	35,053 (26%)	38,629 (26%)		478 (37%)	430 (35%)	
Fair /Poor	12,017 (9.0%)	14,770 (10%)		284 (22%)	335 (27%)	
Inpatient for Mental Health Trt (<12months)			<0.001			0.3
Yes	950 (0.7%)	1,343 (0.9%)	<0.001	109 (8.4%)	118 (9.6%)	<0.001
No	132,417 (99%)	145,544 (99%)		1,181 (92%)	1,106 (90%)	
Mental Illness (<12months)	16,733 (13%)	29,766 (20%)	<0.001	691 (54%)	762 (62%)	<0.001
Major Depressive Episode Lifetime			<0.001			<0.001
Yes	11,610 (8.7%)	22,042 (15%)	<0.001	401 (31%)	523 (43%)	<0.001
No	121,757 (91%)	124,845 (85%)		889 (69%)	701 (57%)	
Major Depressive Episode (<12months)			<0.001			<0.001
Yes	6,231 (4.7%)	12,592 (8.6%)	<0.001	296 (23%)	411 (34%)	<0.001
No	127,136 (95%)	134,295 (91%)		994 (77%)	813 (66%)	
¹ n (%)						
² Pearson's Chi-squared test						

Table 6: Reasons for using prescription pain relievers not as directed by the doctor vs Opioid Use Disorder status, stratified by gender

Characteristic: Reasons for use of pain reliever not as directed by Doctor in the past 12 months	No Opioid Use Disorder, N=280254			Opioid Use Disorder, N=2514		
	1. N = 133,367 ¹	2. N = 146,887 ¹	p-value ²	1. N = 1,290 ¹	2. N = 1,224 ¹	p-value ²
Misuse reason for Pain			0.73			0.11
Yes	3,462 (2.6%)	3,783 (2.6%)		639 (50%)	645 (53%)	
No	129,905 (97%)	143,104 (97%)		651 (50%)	579 (47%)	
Misuse reason To Relax			<0.001			0.048
Yes	1,387 (1.0%)	1,317 (0.9%)		393 (30%)	418 (34%)	
No	131,980 (99%)	145,570 (99%)		897 (70%)	806 (66%)	
Misuse reason To Experiment			<0.001			0.005
Yes	555 (0.4%)	389 (0.3%)		60 (4.7%)	89 (7.3%)	
No	132,812 (100%)	146,498 (100%)		1,230 (95%)	1,135 (93%)	
Misuse reason To get High			<0.001			0.77
Yes	1,382 (1.0%)	853 (0.6%)		471 (37%)	440 (36%)	
No	131,985 (99%)	146,034 (99%)		819 (63%)	784 (64%)	
Misuse reason For Emotions			<0.001			<0.001
Yes	408 (0.3%)	616 (0.4%)		256 (20%)	361 (29%)	
No	132,959 (100%)	146,271 (100%)		1,034 (80%)	863 (71%)	
Misuse reason Other Effect			0.018			0.8
Yes	114 (<0.1%)	90 (<0.1%)		79 (6.1%)	72 (5.9%)	
No	133,253 (100%)	146,797 (100%)		1,211 (94%)	1,152 (94%)	
Misuse reason Hooked			0.82			0.017
Yes	42 (<0.1%)	44 (<0.1%)		219 (17%)	166 (14%)	
No	133,325 (100%)	146,843 (100%)		1,071 (83%)	1,058 (86%)	
Misuse reason For Sleep			<0.001			0.079
Yes	661 (0.5%)	912 (0.6%)		233 (18%)	255 (21%)	
No	132,706 (100%)	145,975 (99%)		1,057 (82%)	969 (79%)	

¹ n (%)

² Pearson's Chi-squared test

3.2.1 PCA with multilinear regression model and evaluation

The PCA identified 47 Principle Components from the 54 column variables in the cleaned NSDUH subset. This represented 0 reduction in dimensionality given 7 were removed at the start of training. Figure 7 and 8 show how the first PC contributes 8% of all the input features variance. By PC36, the cumulative proportion is nearly 90%. When the multilinear regression model was trained on just the first 9 PC's, the R² value was 0.37 (figure 9). As the coefficient of determination ranges from 0 to 1 and a value of 0 indicating that the y variable cannot be explained by the x variables at all, and 1 indicating that y can be perfectly explained without error by x, this shows that 37% of the variance in OUD status can be explained by the 9 principle components here. Out of interest, this was repeated but with 36 PCs, and the R² value changed to 0.42%. The model appears to have significant coefficients for all the principal components (PC1 to PC9) as indicated by the very small p-values (< 2e-16). Finally, with a Mean Absolute Error of 0.015 shows a very close prediction to the actual y variable (fig10)

3.2.2 Classification Model training and evaluation

In google colab, the data was similarly split 80:20 train: test, dropping the y variable from the data and the four heroin and pain reliever abuse and dependence variables as before. The models were tested without PCA and then with PCA. The accuracy, precision, recall, specificity, error rate and F1 scores can be found in Table 7, and the confusion matrices for all 5 models available in Table 8.

Figure 7: graphical representation of pca findings

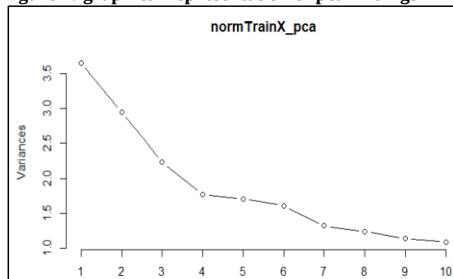


Figure 8: Table of PCA results

```
ksdunpca <- normTrainX_pca$X
# View the summary of the PCA model
summary(normTrainX_pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	1.91256	1.71607	1.4973	1.33339	1.3081	1.27163	1.15133	1.11622	1.06963	1.0443	1.03163	1.02913	1.02073	1.01261
Proportion of Variance	0.07783	0.06266	0.0477	0.03783	0.0364	0.03441	0.02821	0.02651	0.02434	0.0232	0.02264	0.02253	0.02217	0.02182
Cumulative Proportion	0.07783	0.14048	0.1882	0.22601	0.2624	0.29682	0.32504	0.35155	0.37589	0.3991	0.42174	0.44427	0.46644	0.48826

	PC15	PC16	PC17	PC18	PC19	PC20	PC21	PC22	PC23	PC24	PC25	PC26	PC27
Standard deviation	1.00667	1.00439	0.99894	0.99631	0.99083	0.98854	0.98042	0.97532	0.96939	0.96774	0.94516	0.93596	0.93088
Proportion of Variance	0.02156	0.02146	0.02123	0.02112	0.02089	0.02079	0.02045	0.02024	0.01999	0.01993	0.01901	0.01864	0.01844
Cumulative Proportion	0.50982	0.53128	0.55251	0.57363	0.59452	0.61531	0.63576	0.65600	0.67600	0.69592	0.71493	0.73357	0.75201

	PC28	PC29	PC30	PC31	PC32	PC33	PC34	PC35	PC36	PC37	PC38	PC39	PC40	PC41
Standard deviation	0.92941	0.9197	0.91868	0.91577	0.89853	0.8861	0.87065	0.86421	0.8311	0.8112	0.75338	0.74127	0.72573	0.67263
Proportion of Variance	0.01838	0.0180	0.01796	0.01784	0.01718	0.0167	0.01613	0.01589	0.0147	0.0140	0.01208	0.01169	0.01121	0.00963
Cumulative Proportion	0.77039	0.7884	0.80634	0.82418	0.84136	0.8581	0.87419	0.89008	0.9048	0.9188	0.93085	0.94255	0.95375	0.96338

	PC42	PC43	PC44	PC45	PC46	PC47
Standard deviation	0.66554	0.61920	0.57387	0.54752	0.51554	6.077e-15
Proportion of Variance	0.00942	0.00816	0.00701	0.00638	0.00565	0.000e+00
Cumulative Proportion	0.97280	0.98096	0.98797	0.99435	1.00000	1.000e+00

Figure 9: Results of multilinear regression training based on 9 PCs

```
Warning: You are trying to do regression and your outcome only has two possible values Are you
trying to do classification? If so, use a 2 level factor as your outcome column.
Call:
lm(formula = .outcome ~ ., data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-13.0983  -0.0444   0.0038   0.0434  10.8024

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.736e-15  1.663e-03   0.000      1
PC1          2.712e-01  8.693e-04 312.023 < 2e-16 ***
PC2          8.451e-02  9.688e-04  87.226 < 2e-16 ***
PC3          1.302e-01  1.110e-03 117.262 < 2e-16 ***
PC4         -3.084e-02  1.247e-03 -24.732 < 2e-16 ***
PC5          8.919e-03  1.271e-03   7.017 2.27e-12 ***
PC6         -1.028e-01  1.307e-03 -78.592 < 2e-16 ***
PC7         -8.138e-03  1.444e-03  -5.636 1.74e-08 ***
PC8         -1.348e-01  1.489e-03 -90.529 < 2e-16 ***
PC9         -6.554e-02  1.554e-03 -42.168 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7908 on 226204 degrees of freedom
Multiple R-squared:  0.3747,    Adjusted R-squared:  0.3747
F-statistic: 1.506e+04 on 9 and 226204 DF, p-value: < 2.2e-16
```

Figure 10: Results of Mean Absolute Error of multilinear regression model

```
118 # [K]
119 pcaTestX <- predict(normTrainX_pca, normTestX)
120 predictions <- predict(model,pcaTestX[,1:9])
121 normTestY$predictions <- predictions
122 predictedOUD <- back_preProc(yPreprocessor, normTestY)
123 MAE(pred = predictedOUD$predictions, obs = predictedOUD$udpyopi)
124 # [1] 0.01544035
```

Table 7: Evaluation of five classification models

Model: GaussianNB()	Model: BernoulliNB()
Accuracy: 0.98 Precision: 0.33 Recall: 0.81 Specificity: 0.98 Error Rate: 0.02 F1 Score: 0.47	Accuracy: 0.99 Precision: 0.50 Recall: 0.69 Specificity: 0.99 Error Rate: 0.0087 F1 Score: 0.58
Model: DecisionTreeClassifier()	Model: RandomForestClassifier()
Accuracy: 0.99 Precision: 0.74 Recall: 0.79 Specificity: 0.99 Error Rate: 0.0041 F1 Score: 0.77	Accuracy: 0.99 Precision: 0.91 Recall: 0.80 Specificity: 0.999 Error Rate: 0.0024 F1 Score: 0.85
Model: SVC()	<i>Intentionally empty</i>
Accuracy: 0.99 Precision: 0.0 Recall: 0.0 Specificity: 1.0 Error Rate: 0.0086 F1 Score: 0.0	

Table 8: Grid of five Classification model confusion matrices.

PREDICTED	GNB	ACTUAL		PREDICTED	DTC	ACTUAL	
		No OUD	Yes OUD			No OUD	Yes OUD
	No OUD	55252 <i>(True Negative)</i>	812 <i>(False Negative)</i>		No OUD	55930 <i>(True Negative)</i>	134 <i>(False Negative)</i>
	Yes OUD	92 <i>(False Positive)</i>	398 <i>(True Positive)</i>		Yes OUD	100 <i>(False Positive)</i>	390 <i>(True Positive)</i>
PREDICTED	BNB	ACTUAL		PREDICTED	RFC	ACTUAL	
		No OUD	Yes OUD			No OUD	Yes OUD
	No OUD	55721 <i>(True Negative)</i>	343 <i>(False Negative)</i>		No OUD	56028 <i>(True Negative)</i>	36 <i>(False Negative)</i>
	Yes OUD	151 <i>(False Positive)</i>	339 <i>(True Positive)</i>		Yes OUD	99 <i>(False Positive)</i>	391 <i>(True Positive)</i>
PREDICTED	SVC	ACTUAL					
		No OUD	Yes OUD				
	No OUD	56064 <i>(True Negative)</i>	0 <i>(False Negative)</i>				
	Yes OUD	490 <i>(False Positive)</i>	0 <i>(True Positive)</i>				

3.3. Discussions

Huge efforts are being made to understand what illicit substances are being abused the most and who is most affected in American Society. The results of this EDA alone help identify those who are more likely to be OUD, what other substances (either abused on dependent on) correlate with OUD. Poor mental health was also associated with OUD, and key reasons for taking pain relievers in a way that was not as directed by the doctor, were for inadequate pain relief and for the euphoric high. Pharmaceutical companies could therefore focus efforts on developing pain relievers with a better pain relief profile and less of a euphoric sensation.

This data is so important to the many stakeholders mentioned at the start of this report, that there are dedicated websites of compiled reports, interactive charts and data visualizations (34). It is also now possible to explore this NSDUH survey data connected and blended with several other data sources such as Mental Health Client level data, Treatment Episode Data Set: Admissions and Discharges, without having to download any dataset files or additional tools. Outcomes can be explored across different variables to examine association between them by using Public-Use and Restricted Use Data Analysis Systems (PDAS and RDAS)(34).

It is therefore conclusive that exploratory data analysis of the NSDUH is of great importance to stakeholders and will continue to be for the foreseeable future.

However there are some limitations to this dataset. The PUF were heavily imputed already before being released, to avoid potential re-identification of responders. The language availability of just Spanish or English does limit the generalisability. Finally, the survey itself does not include military personnel, those in jail or in nursing homes or institutions, and excludes the homeless. Their absence may have an impact on some of the mental health measures and use of illicit substances. With these being some of the most vulnerable members of a population, it is unfortunate they are not in the dataset.

With respect to creating the machine learning models, the data was heavily skewed to non-OUD, with only 0.89% incidence of OUD. Consequently, most of the classification models performed better, except for SVC, than the regression model. At the time of submission, model performance was not evaluated in terms of Area under the receiver operating characteristics curve (AUROC) or the area under the precision-recall curve (AUPRC) and is a remaining item to perform. The variables that the models were trained on were informed by literature review, and it is possible that other important predictors were omitted, or maybe not even measured by the NSDUH survey. Whilst these classification models could never be used to infer opioid use disorder on an individual level bases (which is not possible given the data is cross-sectional) they may have a place in helping health experts understand from a population level, how best to allocate resources and further research so that steps can be taken earlier to avoid OUD.

4. References

1. CDC Director's Media Statement on U.S. Life Expectancy [press release]. <https://www.cdc.gov/media/releases/2018/s1129-US-life-expectancy.html>, 29th November 2018 2018.
2. Mortality in the United States, 2017 [press release]. <https://www.cdc.gov/nchs/products/databriefs/db328.htm>: CDC, November 2018 2018.
3. Spencer MR MA, Warner M. Drug overdose deaths in the United States, 2001–2021. Hyattsville, MD: National Center for Health Statistics; 2022 December 2022.
4. Hedegaard H MA, Warner M. Drug overdose deaths in the United States, 1999–2019. Hyattsville, MD: National Centre for Health Statistics; 2020 December 2020.
5. U.S. Department of Health and Human Services (HHS) OotSG. Facing Addiction in America: The Surgeon General's Spotlight on Opioids. Washington DC: HHS; 2018 September 2018.
6. Krantz MJ, Palmer RB, Haigney MCP. Cardiovascular Complications of Opioid Use. *Journal of the American College of Cardiology*. 2021;77(2):205-23.
7. Wiese AD, Griffin MR, Schaffner W, Stein CM, Greevy RA, Mitchel Jr EF, et al. Long-acting opioid use and the risk of serious infections: a retrospective cohort study. *Clinical Infectious Diseases*. 2019;68(11):1862-9.
8. Johnson AJ, Jones CW. Opioid use disorders and pregnancy. *Obstetrics and Gynecology Clinics*. 2018;45(2):201-16.
9. Group M. Economic Cost of Substance Abuse Disorder in the United States, 2019 Recovery Centers of America 2201 Renaissance Boulevard, 3rd floor, King of Prussia, PA 19406: Recovery Centres of America; April 2020.
10. Determination that a Public Health Emergency Exists [press release]. 26Oct2017 2017.
11. Initiative NioHH. Fiscal Years 2018–2021 Budget for the NIH HEAL Initiative 2022 [Available from: <https://heal.nih.gov/about/budget>.
12. Quality CfBHSa. 2019 National Survey on Drug Use and Health: Methodological summary and definitions. Rockville, MD: Substance Abuse and Mental Health Services Administration.; 2020.
13. Volkow ND, Blanco C. Interventions to Address the Opioid Crisis-Modeling Predictions and Consequences of Inaction. *JAMA Netw Open*. 2021;4(2):e2037385.
14. Lin DH, Jones CM, Compton WM, Heyward J, Losby JL, Murimi IB, et al. Prescription Drug Coverage for Treatment of Low Back Pain Among US Medicaid, Medicare Advantage, and Commercial Insurers. *JAMA Network Open*. 2018;1(2):e180235-e.
15. Gallamoza B. NSDUH_2015–2019.csv. <https://www.kaggle.com/datasets/bgallamoza/national-survey-of-drug-use-and-health-201520192021>.
16. National Survey on Drug Use and Health (NSDUH) Population Data Public Use Files www.samhsa.gov 2019 [2019; Available from: <https://www.datafiles.samhsa.gov/dataset/national-survey-drug-use-and-health-2019-nsduh-2019-ds0001>.
17. SAMHSA. National Survey On Drug Use And Health: 2014 And 2015 Redesign Changes www.samhsa.gov/2015 [Available from: <https://www.samhsa.gov/data/report/national-survey-drug-use-and-health-2014-and-2015-redesign-changes>.
18. Gong J, Merry AF, Beyene KA, Campbell D, Frampton C, Jones P, et al. Persistent opioid use and opioid-related harm after hospital admissions for surgery and trauma in New Zealand: a population-based cohort study. *BMJ Open*. 2021;11(1):e044493.
19. Compton WM, Jones CM, Baldwin GT. Relationship between Nonmedical Prescription-Opioid Use and Heroin Use. *New England Journal of Medicine*. 2016;374(2):154-63.
20. R. Lawrence DMaLC. Systematic review to determine which validated measurement tools can be used to assess risk of problematic analgesic use in patients with chronic pain. *British Journal of Anaesthesia*. 2017;119:1092-109.
21. Bzdok D, Altman N, Krzywinski M. Statistics versus machine learning. *Nat Methods*. 2018;15(4):233-4.

22. Sauer J, Stewart K. Geographic information science and the United States opioid overdose crisis: A scoping review of methods, scales, and application areas. *Soc Sci Med*. 2023;317:115525.
23. Liu YS, Kiyang L, Hayward J, Zhang Y, Metes D, Wang M, et al. Individualized Prospective Prediction of Opioid Use Disorder. *The Canadian Journal of Psychiatry*. 2023;68(1):54-63.
24. Han D-H, Lee S, Seo D-C. Using machine learning to predict opioid misuse among U.S. adolescents. *Preventive Medicine*. 2020;130:105886.
25. Gao W, Leighton C, Chen Y, Jones J, Mistry P. Predicting opioid use disorder and associated risk factors in a Medicaid managed care population. *Am J Manag Care*. 2021;27(4):148-54.
26. Orgera K TJ. The Opioid Epidemic and Medicaid's Role in Facilitating Access to Treatment. Kaiser Family Foundation 2019.
27. Artificial Intelligence Risk Management Framework (AI RMF 1.0). In: Commerce NIOsTUDo, editor. <https://www.nist.gov/artificial-intelligence2023>.
28. Australia's AI Ethics Principles Australian Government Department of Industry, Science and Resources 2019 [updated 7 November 2019. Available from: <https://www.industry.gov.au/publications/australias-artificial-intelligence-ethics-framework/australias-ai-ethics-principles>.
29. SAMHSA. Results from the 2019 National Survey on Drug Use and Health: Detailed Tables 2019 [26 April 2023]. Available from: <https://www.samhsa.gov/data/sites/default/files/reports/rpt29394/NSDUHDetailedTabs2019/NSDUHDetTa bsAppA2019.htm>.
30. SAMHSA. 2021 National Survey of Drug Use and Health (NSDUH) Releases 2021 [Available from: <https://www.samhsa.gov/data/release/2021-national-survey-drug-use-and-health-nsduh-releases>.
31. Jain K, Singh A, Singh P, Yadav S. An Improved Supervised Classification Algorithm in Healthcare Diagnostics for Predicting Opioid Habit Disorder. *International Journal of Reliable and Quality E-Healthcare*. 2022;11(1):1-16.
32. Davis J, Goadrich M. The relationship between Precision-Recall and ROC curves. *Proceedings of the 23rd international conference on Machine learning*; Pittsburgh, Pennsylvania, USA: Association for Computing Machinery; 2006. p. 233–40.
33. Saito T, Rehmsmeier M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PloS one*. 2015;10(3):e0118432.
34. SAMHSA. Analyze Data 2023 [Available from: <https://www.datafiles.samhsa.gov/analyze-data>.



Appendix

Appendix A: Appendix A EDA of NSDUH 2015_2019.Rmd
Source code for EDA

Appendix B: Model Markdown.Rmd
Source code for Multiple Linear Regression model and testing

Appendix C: Classification Models for NSDUH

- PDF file of source code in google colab: multiple classification model testing evaluating.pdf
- ipynb file of source code for google colab: RerunClassifications.ipynb

Appendix A to C provided in zip file names SID Source Code. Appendix D:

Table 1 of variables of interest that formed the NSDUH_id Subset.

Appendix D

Table 1: Variables of interest that formed the NSDUH_id subset

Theme	Code	Question	Variable for Machine Learning	Responses	Cleaning required?
Dependent Variable	UDPYOPI	Recoded OPIOID DEPENDENCE OR ABUSE - PAST YEAR	y	0 = No (ABODHER=0 and UDPYPNR=0).	NIL
				1 = Yes (ABODHER=1 or UDPYPNR=1)	
Demographic/socioeconomic/geographic: 10 variables	year	Added into Kaggle set	N/A: just for EDA train the model by it	2015, 2016, 2017,2018,2019	NIL
	id	Added into Kaggle set		1:287600	NIL
	irsex	IMPUTATION revised gender	x	1= male	NIL
				2=female	
	catag6	AGE CATEGORY RECODE	x	1 = 12-17 Years Old	NIL
				2 = 18-25 Years Old	
				3 = 26-34 Years Old	
				4 = 35-49 Years Old	
				5 = 50-64 Years Old	
	newrace2	RACE/HISPANICITY RECODE (7 LEVELS)	x	6 = 65 or Older	NIL
				1 = NonHispanic White	
				2 = NonHispanic Black/African American	
				3 = NonHispanic Native American/Alaska Native	
				4 = NonHispanic Native Hawaiian/Other Pacific Islander	
	eduhighcat	EDUCATION CATEGORIES	x	5 = NonHispanic Asian	NIL
				6 = NonHispanic more than one	
				7 = Hispanic	
				1 = Less than high school	
				2 = High school grad	
	irwrkstat	EMPLOYMENT STATUS IMPUTED REVISED	x	3 = Some college/Associate degree	NIL
				4 = College graduate	
				5 = 12 to 17 year olds	
				1 = Employed full time	
				2 = Employed part time	
	income	RC-TOTAL FAMILY INCOME RECODE	x	3 = Unemployed	NIL
				4 = Other (incl. not in labor force)	
				99 = 12-14 year	
				1 = Less than \$20,000	
	irinsur4	OVERALL HEALTH INSURANCE	x	2 = \$20,000 - \$49,999.	NIL
				3 = \$50,000 - \$74,999	
	govtprog	RC-PARTICIPATED IN ONE OR MORE GOVT ASSIST PROGRAMS	x	4 = \$75,000 or MORE	NIL
				1= yes	
	coutyp4	COUNTY/METRO/NON METRO STATUS	x	2=No	NIL
				1 = Large Metro Nonmetro	
				2 = Small Metro	
Other Health variables: 5 variables	health2	OVERALL HEALTH RECODE	x	3 = Nonmetro	NIL
				1= Excellent	
				2= very good	
				3= Good	
				4= Fair/Poor	
	amhinp2	RECEIVED INPATIENT MENTAL HEALTH TREATMENT IN PAST YR	x	. = unknown.	impute to median
				1= yes	NIL
				2=No	
	amiyr_u,	The following variable, AMIYR_U, is an indicator for Any Mental Illness (AMI) based on the 2012 revised predicted probability of SMI (SMIPP_U).	x	. = unknown.	impute to median
				0=no in past year	NIL
				1= yes.	
	amdelt,		x	. = 12-17yrs	NIL
				1= yes	

		Recorded major depressive episode in their lifetime		2= no .= 12-17yrs/unknown		
	amdeyr.	Recorded major depressive episode in past year	x	0=no in past year 1= yes. .= 12-17yrs	NIL	
	abusepypnr	recoded abuse of pain reliever.	x (strong indicator to y)	0=no/unknown otherwise. 1=yes	NIL	
	abusepysed,	recoded abuse sedative	x	0=no/unknown otherwise. 1=yes	NIL	
Past 12 months prior Substance Abuse variables: 12 variables	abusepypsy.	pyschotherapeutic	x	0=no/unknown otherwise. 1=yes	NIL	
	abusealc	alcohol	x	0=no/unknown otherwise. 1=yes	NIL	
	abusepytrq,	tranquilizer	x	0=no/unknown otherwise. 1=yes	NIL	
	abusepystm,	stimulant	x	0=no/unknown otherwise. 1=yes	NIL	
	abusemrj,	marijuana	x	0=no/unknown otherwise. 1=yes	NIL	
	abusecoc,	cocaine	x	0=no/unknown otherwise. 1=yes	NIL	
	abuseher,	heroin	x (strong indicator to y)	0=no/unknown otherwise. 1=yes	NIL	
	abusepyhal	hallucinogen	x	0=no/unknown otherwise. 1=yes	NIL	
	abusepyinh	inhallant	x	0=no/unknown otherwise. 1=yes	NIL	
	abusepymth,	methamphetamine	x	0=no/unknown otherwise. 1=yes	NIL	
	depndalc	alcohol	x	0=no/unknown otherwise. 1=yes	NIL	
Prior 12 months Substance Dependence variables: 13 variables	depndmrj	marijuana	x	0=no/unknown otherwise. 1=yes	NIL	
	depndcoc	cocaine	x	0=no/unknown otherwise. 1=yes	NIL	
	depndher	heroin	x (strong indicator to y)	0=no/unknown otherwise. 1=yes	NIL	
	depndpyhal	hallucinogen	x	0=no/unknown otherwise. 1=yes	NIL	
	depndpyinh	inhallant	x	0=no/unknown otherwise. 1=yes	NIL	
	depndpymth",	methamphetamine	x	0=no/unknown otherwise. 1=yes	NIL	
	depndpypnr,	recoded dependence of pain reliever.	x (strong indicator to y)	0=no/unknown otherwise. 1=yes	NIL	
	depndpytrq,	tranquilizer	x	0=no/unknown otherwise. 1=yes	NIL	
	depndpystm,	stimulant	x	0=no/unknown otherwise. 1=yes	NIL	
	depndpysed,	sedative	x	0=no/unknown otherwise. 1=yes	NIL	
	depndpypsy,	pyschotherapeutic	x	0=no/unknown otherwise. 1=yes	NIL	
	"ndssdnsp".	Nicotine	x	0=no/unknown otherwise. 1=yes	NIL	
Reasons given for misusing Prescription Pain Relievers variables: 8 variables	pnrrsrelx, pnrrshigh, pnrrslep, pnrrse xpt, pnrrspain, pnrrsemot, pnrrsd gfx, pnrrshook	USED LAST PAIN RLVR (NOT DIRECTED) TO: RELAX, GET HIGH, SLEEP, EXPERIMENT, PAIN RELIEF, EMOTIONS, OTHER EFFECTS, HOOKED	x	1= yes, 2=No. 3=yes logically assigned 83=did not use pain reliever in past 12 months. 85= bad data 91= never used/misused pain reliever 93= did not misuse pain reliever in past 12 months 94=don't know 97= refused 98=blank	nil nil recode to 1=yes recode to 2=no recode to 2=no recode to 2=no recode to 2=no recode to 2=no recode to 2=no recode to 2=no	